

TwinLoc

Digital Twin Based Wireless Sensor Network
Localization
using Hybrid MFO-GA

Experimental Report

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Project: TwinLoc - Digital Twin Based Wireless Sensor Network Localization
Algorithms: MFO, GA, Hybrid MFO-GA
Experimental Runs: 30
Network Size: 50 Sensor Nodes
Anchor Nodes: 10
Communication Range: 100 m

Executive Summary

This report presents the comparative performance evaluation of three localization algorithms: MFO, GA and Hybrid MFO-GA for Wireless Sensor Network (WSN) localization. Thirty independent simulation runs were conducted to evaluate Mean Localization Error (MLE), Root Mean Square Error (RMSE), Normalized Localization Error (NLE), and Localization Success Rate. The results provide a statistical comparison of localization accuracy and algorithm consistency under identical experimental conditions.

Experiment Configuration

Parameter	Value
Experimental Runs	30
Sensor Nodes	50
Anchor Nodes	10
Communication Range	100 m
Algorithms	MFO GA Hybrid MFO-GA

Table 1. Statistical Comparison of Localization Algorithms

Algorithm	Avg MLE	Std MLE	Avg RMSE	Std RMSE	Avg NLE	Avg Success Rate
MFO	214.299	22.2247	252.2518	25.1329	2.143	100.0
GA	142.7652	17.2898	185.2965	25.6276	1.4277	100.0
Hybrid MFO-GA	160.023	23.8666	213.2231	32.8158	1.6002	100.0

Statistical Comparison Graphs

Figure 1. Average Mean Localization Error Comparison

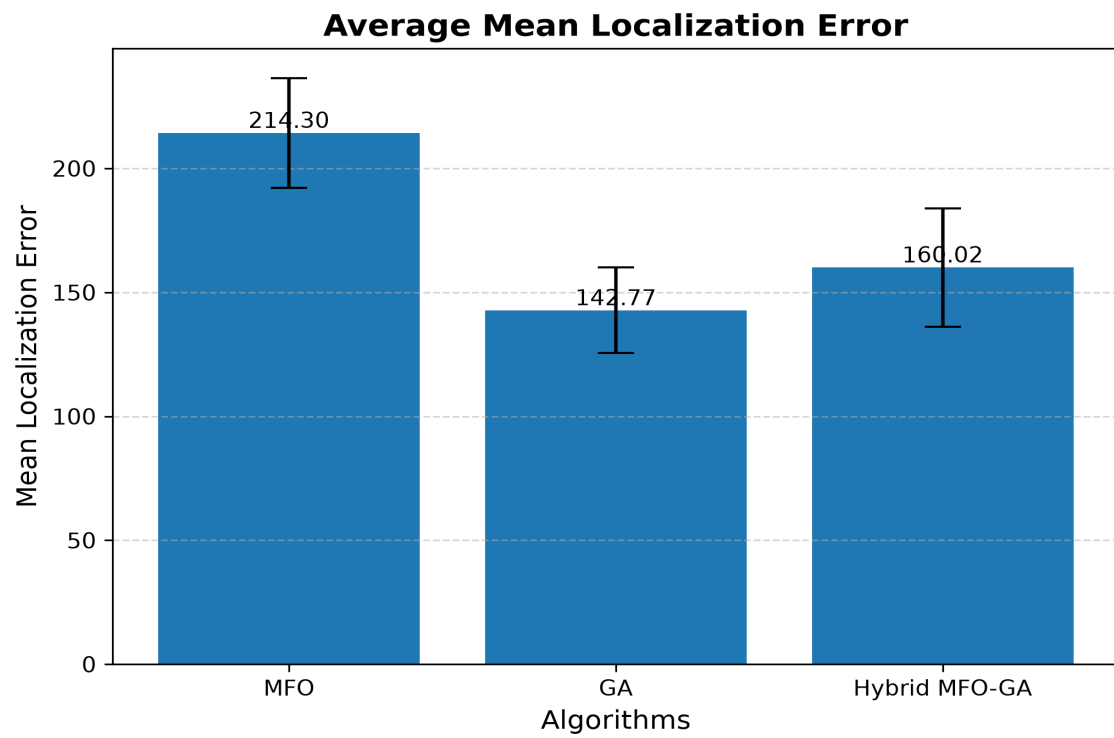


Figure 2. Average Root Mean Square Error Comparison

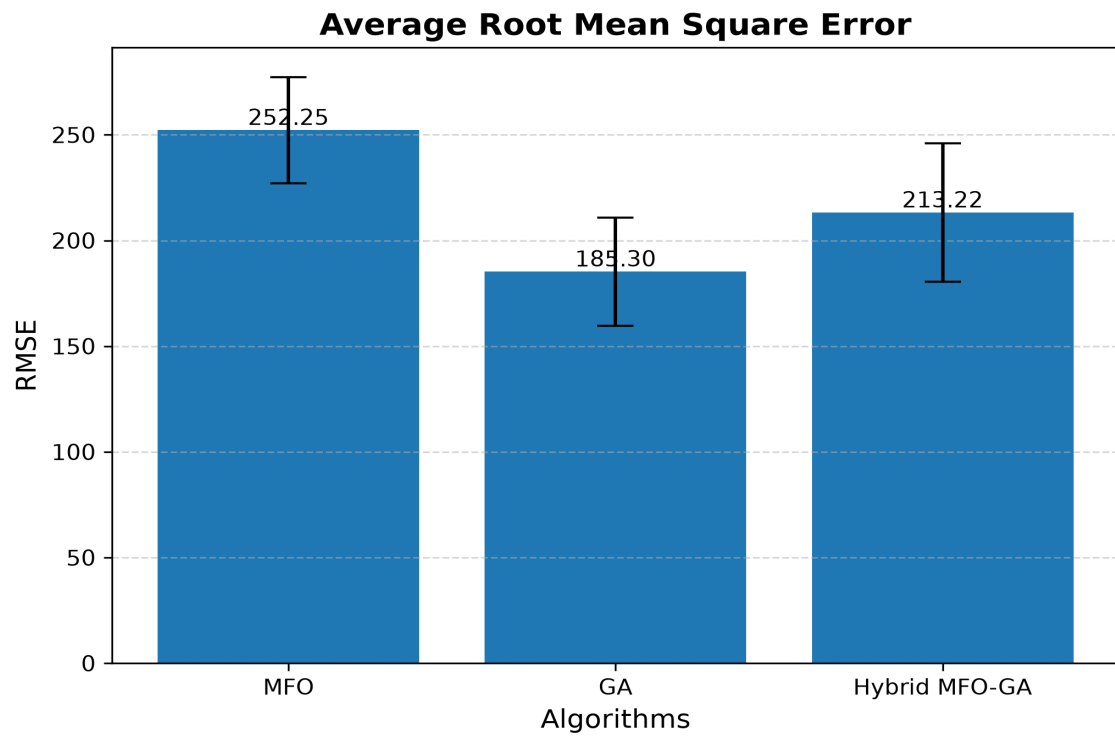


Figure 3. Average Normalized Localization Error Comparison

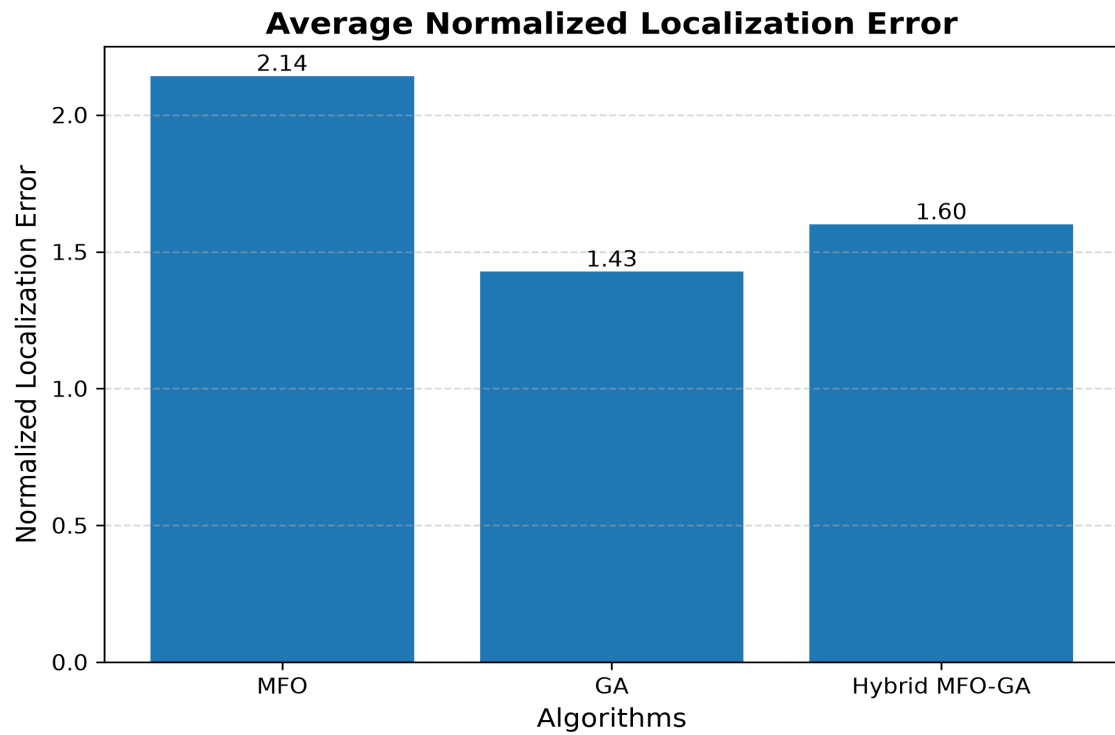
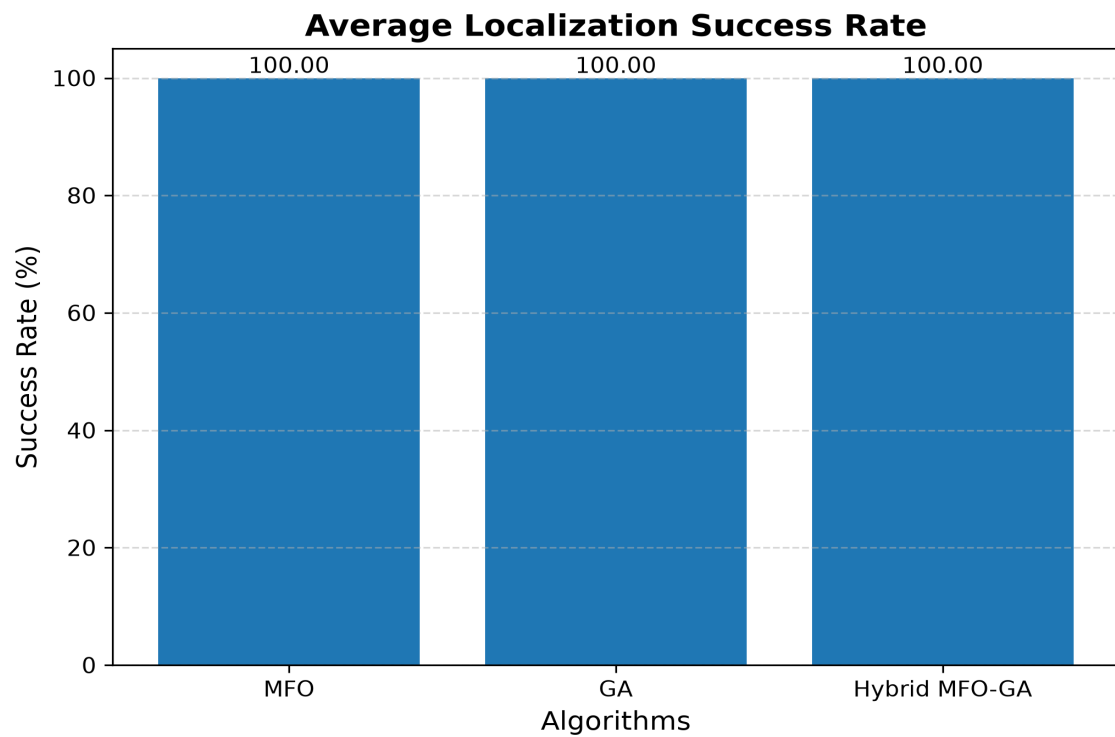


Figure 4. Average Localization Success Rate Comparison



Experimental Conclusions

1. Best Average RMSE

GA achieved the lowest Average RMSE of **185.30**.

2. Best Mean Localization Error

GA achieved the lowest Average MLE of **142.77**.

3. Lowest Normalized Localization Error

GA achieved the lowest Average NLE of **1.43**.

4. Highest Localization Success Rate

All evaluated algorithms achieved 100.00% localization success.

5. Most Consistent Algorithm

MFO produced the lowest RMSE standard deviation of **25.13**.

Overall Observation

Based on the statistical evaluation of 30 independent simulation runs, GA demonstrated the best overall localization performance by achieving the lowest Mean Localization Error (MLE), Root Mean Square Error (RMSE), and Normalized Localization Error (NLE), while all evaluated algorithms maintained a 100% localization success rate. These results indicate that GA is the most effective algorithm among the evaluated methods for the current wireless sensor network configuration.

Future Work

- Integrate PSO, GWO, TLBO and BAT localization algorithms.
- Evaluate larger wireless sensor networks containing hundreds of nodes.
- Extend TwinLoc with real-time Digital Twin visualization.
- Compare Hybrid MFO-GA with modern optimization techniques.
- Deploy TwinLoc as a cloud-based localization platform.